

Related work: digital mice in context

Executive summary

The closest antecedents to “digital mice” fall into five threads. First, recent work explicitly reframes cognitive bias and bounded rationality as *instruments* for computational social science rather than defects to mitigate (CoBRA; CoBRA, 2025), and a parallel line uses LLMs as “Homo silicus” — prompt-conditioned implicit models of humans to be experimented on (Horton, 2023; Manning et al., 2024). Second, the silicon-sampling tradition (Argyle et al., 2023) and its digital-twin descendants (Peng et al., 2025) build synthetic populations through demographic persona conditioning, but typically without manipulating reasoning capacity. Third, an emerging auctions-and-games literature deploys LLMs as economic agents and benchmarks their behavior against the canonical human experimental record (Shah et al., 2025; Zheng et al., 2025; Henning et al., 2025), in some cases formalizing bounded rationality through QRE (Pechon-Elkins and Chun, 2026) or prompt-level equilibrium concepts (Zhu, 2025). Fourth, Li (2017)-style OSP theory (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022) provides the diagnostic vocabulary — obvious dominance, contingent reasoning failures — that the digital-mice fitness function reuses. Fifth, multiple empirical studies document that off-the-shelf LLMs are *too* rational, too homogeneous, and too fair relative to humans (Henning et al., 2025; Peng et al., 2025), which is precisely the gap a controlled-deficit pipeline is built to close.

Digital mice are novel along two axes that no prior line covers jointly: (i) systematic, prompt-induced reasoning-capacity knockouts treated as an experimental design variable (a “knockout mouse” for LLM agents), and (ii) quantitative calibration of those knockouts against Li (2017) auction moments, then used as the fitness signal in an evolutionary search over auction genotypes. Prior work has the ingredients — persona conditioning, bias toolkits, auction benchmarks, QRE calibration — but none combines instructed cognitive deficits with moment-matched human auction data inside a mechanism-design search loop.

1. Bias and bounded rationality as research instruments

A small but growing literature reframes LLM cognitive limitations as something to *program in*, not engineer away. CoBRA (CoBRA, 2025) provides a toolkit with a Cognitive Bias Index and a Behavioral Regulation Engine that explicitly controls biases in LLM-based social agents, using validated social-science experiments as the scaffold, and positions “bias not as a flaw but as an instrument for scientific reproducibility and behavioral modeling.” This is the same instrumental move the digital-mice analogy makes: treat the deficit as the apparatus, not the noise. The contrast is along the deficit axis — CoBRA targets classical cognitive biases (anchoring, framing, etc.), whereas digital mice target *reasoning depth and contingent reasoning capacity* relevant to auction strategy. Neither library calibrates against quantitative auction moments.

Zheng et al. (2025) push in the same direction empirically: across repeated games (RPS, Prisoner’s Dilemma) LLMs reproduce familiar human heuristics such as outcome-based strategy switching and increased cooperation under repeated interaction, but “apply these rules more rigidly and demonstrate weaker sensitivity to the dynamic changes in the game environment.” They conclude that current LLMs “capture only a partial form of human-like bounded rationality,” motivating training methods for flexible opponent modeling. Digital mice take the opposite engineering route: rather than retraining for more flexible bounded rationality, we *prompt-instrument* specific deficits and treat the resulting behavior as the unit of analysis.

2. Homo silicus and automated social science

Horton (2023) introduces the “Homo silicus” framing: LLMs as “implicit computational models of humans ... they can be given endowments, information, preferences, and so on, and then their behavior can be explored in scenarios via simulation.” Methodologically, the paper replicates Charness–Rabin, Kahneman et al., Samuelson–Zeckhauser, Oprea, and Horton himself, and treats *qualitative* agreement (or generative divergence) as the validation criterion. Digital mice keep the prompt-as-endowment idea but replace qualitative replication with quantitative moment matching against Li (2017).

Manning et al. (2024) push this to the full research loop: LLMs as both *scientists* (hypothesis generators) and *subjects* (experimental agents), with structural causal models serving as the language for stating hypotheses, building agents, designing experiments, and analyzing data, demonstrated across negotiations, bail hearings, job interviews, and auctions. Crucially, the paper does not introduce controlled cognitive deficits or bounded-rationality constraints — agents are treated as default reasoners. Digital mice can be read as a specialization of this framework where the SCM “treatment” variable is reasoning capacity itself, instrumented via prompted CoT suppression.

3. Synthetic populations and digital twins

Argyle et al. (2023) coin “silicon sampling” and the criterion of “algorithmic fidelity”: conditioning GPT-3 on thousands of sociodemographic backstories from real survey respondents, with the goal that “proper conditioning will cause it to accurately emulate response distributions from a wide variety of human subgroups.” The construct propagates downstream into the digital-twin literature. Methodologically, this is persona/demographic conditioning, with no explicit treatment of bounded rationality or reasoning-capacity ablation — a clear contrast with the digital-mice approach, which holds persona fixed and varies cognitive capacity.

Peng et al. (2025) stress-test the digital-twin program across nineteen pre-registered replications (164 outcomes) and report an average correlation of only $r \approx 0.20$ with human responses, with documented distortions including a “Targeting Fairness” regime in which LLMs produce systematically skewed fairness judgments in economic and social scenarios. The finding is directly relevant to LLMs-as-bidders or LLMs-as-matching-market participants, where fairness and envy considerations shape bid shading and preference reports. Digital mice respond to this critique by refusing the naive “proxy” framing: rather than asking whether LLMs *are* humans, we ask whether instructed-deficit LLMs can reproduce the *moments* of human play under specific cognitive constraints.

4. LLMs as auction and game-theoretic agents

Shah et al. (2025) deploys LLMs as bidders in canonical experimental auction frameworks (first-price, second-price, and OSP variants), running over a thousand auctions and reporting that LLM bidders “produce results consistent with risk-averse human bidders” and “succumb to the winner’s curse in common value settings.” The paper explicitly engages Li (2017) on obviously strategy-proof mechanisms and Breitmöser and Schweighofer-Kodritsch (2022) on “obviousness” as a theoretical lens for LLM cognitive limitations, and quantitatively benchmarks against Coppinger et al. (1980), Cox et al. (1988), and Kagel (1993) — e.g., matching “the 27% reported by Kagel (1993)” for value-bidding frequency in SPSB. This is the most direct precedent for the auction-side

empirical loop in digital mice, sharing both the OSP diagnostic language and the human-baseline calibration target. The key difference: Shah et al. (2025) does *not* systematically manipulate reasoning capacity to recover specific moments, and does not run an evolutionary search over auction formats conditional on agent cognition.

Pechon-Elkins and Chun (2026) provide a complementary calibration tool by estimating Quantal Response Equilibrium λ parameters for LLM agents on a continuous scale against human data (with human $\lambda \in [1.0, 2.5]$ and frontier-model estimates in $[0.05, 1.10]$ across 1,855 games and seven models). QRE gives a principled one-dimensional handle on strategic sophistication, and is in principle complementary to digital mice: QRE measures *how* rational an LLM is, while digital mice manipulate that quantity through instructed deficits.

Zhu (2025) formalizes an “LLM-Nash reasoning equilibrium” where agents choose *reasoning prompts* rather than actions, and defines mindsets as triples (information intake, prompt space, worldview parameters), with equilibrium defined over the prompt space and actions emerging as LLM inference output. This provides a clean theoretical vocabulary for what digital mice do mechanically: deficit-injecting prompts are precisely the strategic objects in $\mathcal{X}, \mathcal{Y}$ above, and the equilibrium concept formalizes why prompt-level manipulations produce behavior distinct from classical Nash play.

5. Out-of-the-box LLMs are too rational and too homogeneous

A robust empirical finding across studies is that frontier LLMs, absent intervention, are systematically *more* rational and *more* homogeneous than human subjects. Henning et al. (2025) run the Smith–Suchanek–Williams bubble paradigm with LLM populations and find that LLMs “generally exhibit a *textbook-rational* approach, pricing the asset near its fundamental value, and show only a muted tendency toward bubble formation,” and that LLM-based agents “display less trading strategy variance in contrast to humans.” They warn that “LLM-only data . . . to replicate human-driven market phenomena” is unreliable because “key behavioral features, such as large emergent bubbles, were not robustly reproduced.” Together with the fairness distortions documented by Peng et al. (2025) and the rigidity findings of Zheng et al. (2025), this body of evidence motivates the central digital-mice design choice: do not assume a baseline LLM matches a human bidder; instead, calibrate a *cognitive treatment* that reproduces specified moments of human play, and use that treated population as the unit of analysis.

6. Where digital mice are novel vs. reuse

Reused ingredients. The instrumental framing of bias/bounded rationality (CoBRA, 2025); the Homo-silicus prompt-as-endowment paradigm (Horton, 2023; Manning et al., 2024); the LLM-as-auction-bidder setup and OSP diagnostic vocabulary (Shah et al., 2025; Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022); the human-benchmark targets from classical experimental auctions (Coppinger et al., 1980; Cox et al., 1988; Kagel, 1993); and the quantitative measurement of LLM bounded rationality via QRE (Pechon-Elkins and Chun, 2026) or prompt-space equilibrium (Zhu, 2025).

Novel combination. Digital mice (i) treat reasoning-capacity manipulation (CoT suppression, instructed deficits) as a designed experimental variable in the spirit of a knockout mouse, going beyond persona conditioning (Argyle et al., 2023; Peng et al., 2025) and beyond bias libraries

(CoBRA, 2025); (ii) calibrate the induced deficit population against Li (2017) auction moments rather than against qualitative replication (Horton, 2023) or single-dimensional QRE (Pechon-Elkins and Chun, 2026); and (iii) plug the calibrated population into an evolutionary search over auction genotypes with an SMAD-only fitness function — a mechanism-design loop with no antecedent in the LLM-agent literature.

References

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